FARM MANAGEMENT SYSTEM

Project work submitted to Thiruvalluvar University, Serkkadu, Vellore.

In partial fulfilment for the award of degree.

BACHELOR OF SCIENCE

IN

INFORMATION SYSTEM MANAGEMENT

BY

DANUSH S

DEEPA R

DHARMESH S

DHILIP K

GANGABALAN G

DEPARTMENT OF INFORMATION SYSTEM MANAGEMENT SHANMUGA INDUSTRIES ARTS &SCIENCE COLLEGE

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Certified Under Section 2(f) & 12B of the UGC Act 1956,
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Tiruvannamalai – 606603

Instrouction:

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I wanttouseacentralizedSalesforce-basedplatformtorecordandmanage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive informationisprotectedwhileenablingsmoothcollaborationamong users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

1. INTRODUCTION

ProjectOverview:

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

Purpose:

Themainpurposeofthisprojectisto:

- Digitizeagriculturaldatamanagement
- Improvedataaccuracyandaccessibility
- Automaterecordhandling
- Enablereal-timemonitoring
- Supportdata-drivenfarmingdecisions
- Enhanceproductivityandtransparency

2. IDEATIONPHASE

ProblemStatement(CustomerProblemStatementTemplate)

PS	I am (Customer)	I'mtrying to	But	Because	Which makesme feel
PS-1	Farmer	Maintain farm and croprecords	Dataisoften incomplete or incorrect	Recordsare kept manuallyin registers	Frustrated and confused
PS-2	FarmManager	Monitor farm productivity and sales	Reports are delayedand inaccurate	Data is scattered acrossfiles and sheets	Stressedand inefficient
PS-3	Buyer	Accesscrop availability and pricing	Information is not updated regularly	Thereisno centralized digital system	Uncertain and dissatisfied
PS-4	SystemAdmin	Manageuser access and system security	Unauthorize d access may occur	Access controlis handled manuall y	Concerned andinsecure

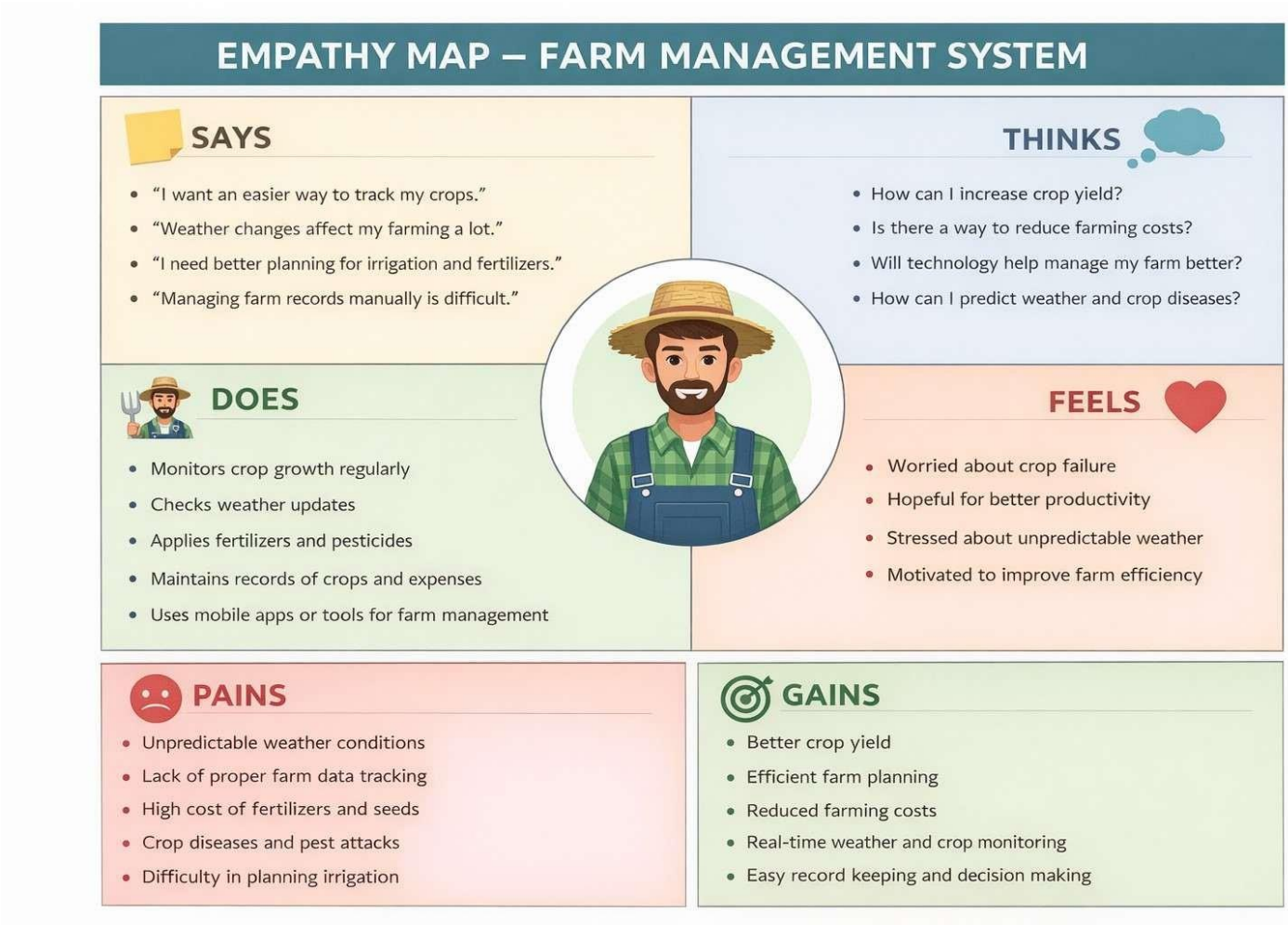
EmpathyMapCanvas

TargetUsers:

- Farmers
- FarmManagers
- Buyers
- SystemAdministrators

EmpathyAnalysis

- **Think&Feel:**Needaccurateandtimelyfarm data
- **Hear:**Feedbackfromagriculturalofficers
- **See:**Registers,paperfiles,spreadsheets
- **Say&Do:**Enterrecordsmanually
- **Pain:**Dataloss,errors,delays
- **Gain:**Automation,fastreporting



Brainstorming

BrainstormedIdeas

- Manualfarmrecords
- Excel-basedmanagement
- Web-basedsystem
- Mobilefarmapp
- Salesforce-basedFMS

SelectedIdea

Salesforce-basedFarmManagementSystem.

Brainstorming Map for Farm Management System

Farm Management System

Possible Features
- ✔ Crop Monitoring
- ✔ Irrigation Scheduling
- ✔ Livestock Tracking
- ✔ Fertilizer Management
- ✔ Weather Updates

Key Problems
- Manual Record Keeping
- Weather Uncertainty
- Crop Monitoring Diffculty
- Market Price Changes
- Resource Wastage

Potential Users
- Farmers
- Farm Owners
- Agricultural Officers
- Farm Workers

Key Benefits
- Better Productivity
- Cost Control
- Improved Decisions
- Time Saving

Farm Resources
- ✔ Land
- ✔ Seeds
- ✔ Fertilizers
- ✔ Water
- ✔ Equipment
- ✔ Labor

Technology
- Mobile App
- Cloud Database
- GPS Tracking
- IoT Sensors

Key Benefits
- ✔ Better Productivity
- ✔ Cost Control
- ✔ Improved Decisions
- ✔ Time Saving

Brainstorming Goal: Generate ideas and features to create a smart **farm management system** that enhances efficiency and productivity for farmers.

3. REQUIREMENTANALYSISPHASE

CustomerJourneyMap

Stages:

1. Login
2. AccessFarmManagementApp
3. EnterFarmandCrop Data
4. RegisterFarmersandBuyers
5. GenerateReports
6. AnalyzeResults
7. DecisionMaking

Customer Journey Map for Farm Management System

Stage	Goals	Actions	Feelings	Touchpoints
1. **Identify Needs**	Determine what is needed for the upcoming season.	The farmer assesses what to plant, required resources like seeds, fertilizers, water, equipment	Curious, **Thoughtful** *considering essential needs*	• Note-taking • Discussions • Weather Forecast
2. **Plan & Organize**	Plan crops, tasks, and resources efficiently.	The farmer uses the Farm Management System to plan crop planting, schedule irrigation, and allocate resources like labor and equipnt.	**Organized, Prepared** *feeling ready for the season*	• Farm Management System dashboard • Planning Tools
3. **Monitor & Track**	Keep track of crop growth, resource usage, and expenses.	The farmer inputs data in the system to monitor crop health, adjust irrigation, track fertilizer use, and record expenses.	**Aware, Informed** *knowing the farm's status*	• Mobile App • Notifications • Weather Updates
4. **Analyze & Decide**	Review data and make informed farming decisions.	The farmer reviews collected data on yields, expenses, and market prices to decide on harvesting and selling crops.	**Confident, Strategic** *making smart, data-driven decisions*	• Reports • Analytics • Market Price Updates
5. **Evaluate & Improve**	Assess results and find areas for improvement.	The farmer evaluates the season's performance and identifies ways to improve crop productivity and reduce costs.	**Reflective, Motivated** *looking to improve for the future*	• Summary Reports • Performance Feedback

Overall Experience: Regular use of the Farm Management System makes farming more organized, efficient, and **profitable.**

SolutionRequirements

FunctionalRequirements

FRNo	Requirement
FR-1	UserAuthentication
FR-2	Farm,Crop,Farmer,BuyerManagement
FR-3	DataEntryandValidation
FR-4	RelationshipMapping
FR-5	ReportGeneration
FR-6	DashboardVisualization
FR-7	UserRoleManagement

Non-FunctionalRequirements

NFR	Description	Example
Usability	Easyinterface	SimpleLightningUI
Security	Dataprotection	Profiles&roles
Reliability	Nodataloss	Cloudbackup

Performance	Fastaccess	Reportsin<3sec
Availability	24/7access	Cloudhosting
Scalability	Handlesgrowth	Morefarms/users

DataFlowDiagram

DataFlow:

User→SalesforceUI→CustomObjects(Farm,Crop,Farmer,Buyer)→Validation Rules → Relationships → Reports → Dashboards

Data Flow Diagram for Farm Management System

INPUTS

- Crop Data
- Resource Data
- Expenses
- Irrigation Schedules
- Worker Schedules
- Activity Reports
- Field Data

Farm Management System

OUTPUTS

- Weather Info
- Cost Reports
- Work Schedules
- Task Instructions
- Crop Advice
- Performance Feedback
- Farm Reports
- Recommendations

INPUTS

- Crop Data
- Irrigation Schedules
- Expenses
- Irrigation Schedules

OUTPUTS

- Weather Info
- Cost Reports
- Work Schedules
- Task Instructions

Small Farmers

- Crop Data
- Irrigation Schedules
- Expenses

Large Farm Owners

- Resource Data
- Worker Schedules
- Expenses
- Activity Reports

Farm Workers

- Task Updates
- Activity Reports

Agricultural Officers

- Field Data
- Current Farm Issues
- Farm Reports
- Recommendations

Customers for Farm Management System (FMS) in a context-level Data **Flow Diagram;** points farm management system to plan crops, manage **resources**, track **farm activities**, and improve productivity.

TechnologyStack

Layer	Technology
Frontend	SalesforceLightningUI
Backend	SalesforcePlatform
Database	SalesforceCloud
Automation	ValidationRules
Analytics	Reports&Dashboards

4. PROJECTDESIGNPHASE

Problem–SolutionFit

The FMS replaces manual farming records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and timely agricultural insights.

ProposedSolution

FunctionalRequirements

Parameter	Description
Problem	Manualmaintenanceoffarm,crop,farmer,andbuyerrecords leading to errors, data loss, and delays.

Solution	Salesforce-basedFarmManagementSystemwithcentralized data storage and automation.
Innovation	Automateddataentryvalidation,cloudstorage,andreal-time reporting dashboards.
Impact	Improveddataaccuracy,fasterreporting,andbetter decision-making for farmers and managers.es
BusinessModel	Digitalfarmmanagementandagriculturaldataservicesplatform.
Scalability	Supportsmultiplefarms,users,andregionswithcloud-based infrastructure.

SolutionArchitecture

ArchitectureLayers

1. PresentationLayer–SalesforceLightningUI
2. ApplicationLayer–ValidationRules,Relationships
3. DataLayer–CustomObjects
4. AnalyticsLayer–ReportsandDashboards

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5. PROJECTPLANNING&SCHEDULING

ProjectPlanning

- CreatedSalesforceDeveloperOrg
- Developedcustomobjects
- ConfiguredtabsandLightningApp
- Createdfieldsandrelationships
- Designedprofilesandroles
- Createdusersandsharingrules
- Developedreportsanddashboards
- Conductedtestingandreviews

ProductBacklog,SprintSchedule,andEstimation(4Marks)

Usethebelowtemplatetocreateproductbacklogandsprintschedule

Sprint	Functional Requirement (Epic)	User Story Number	UserStory/ Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developeraccountsothatIcansetup and deploy the Farm Management System.	3	High	Sanjai E
Sprint-2	DataModeling	USN-2	As an admin, I want to create Farm, Crop, Farmer, and Buyer objects with properrelationshipssothatagricultural data is stored and managedefficiently.	5	High	Simbu S
Sprint-2	DataModeling	USN-3	As a staff member, I want tabs and a LightningAppforallfarmmodulesso that I can easily navigate and perform daily operations.	5	High	Sakthivel K
Sprint-3	Automation	USN-4	Asasystem,Iwanttovalidatefarm and crop details such as dates, quantities, and mandatory fields so that incorrect or incompleterecords areprevented.	3	High	Naveen N
Sprint-3	Automation	USN-5	As a manager, I want automated flows for crop updates and buyer notificationssothatfarmoperations areprocessedfasterwithoutmanual intervention.	3	High	Sanjai E
Sprint-4	Security	USN-6	As an administrator, I wantrole-based access control for farmers,managers,andbuyersso that sensitive farm and transaction data is secure.	5	High	Simbu S
Sprint-5	Reports	USN-7	As a farm manager, I want reports showingcropyield,sales,andbuyer transactions so that I can monitor farm performance and makeinformeddecisions.	4	High	Sakthivel K
Sprint-6	Dashboards	USN-8	As a farm manager, I want dashboards showingcropproductivity,salestrends, and operational statistics so that I can monitor performance visually.	4	Medium	Naveen N

ProjectTracker,Velocity&BurndownChart:(4Marks)

Sprint	TotalStory Points	Duratio n	SprintStart Date	SprintEndDate (Planned)	StoryPoints Completed(as on Planned End Date)	SprintRelease Date (Actual)
Sprint 1	20	6Days	07 Feb2026	13 Feb2026	20	13 Feb2026
Sprint 2	20	6Days	15 Feb2026	21 Feb2026	18	20 Feb2026
Sprint 3	20	6Days	02 Mar2026	08 Mar2026	19	05 Mar2026
Sprint 4	20	6Days	10 Mar2026	16 Mar2026	16	16 Mar2026

6. ProjectDevelopmentPhase:

ProjectDescription:

TheFarmManagementSystem(FMS)inSalesforceisacomprehensivesolution designedtodigitizeandstreamlineagriculturaloperations.Thisapplicationempowers farmers, farmmanagers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

Milestone1-Salesforce:

WhatIsSalesforce?

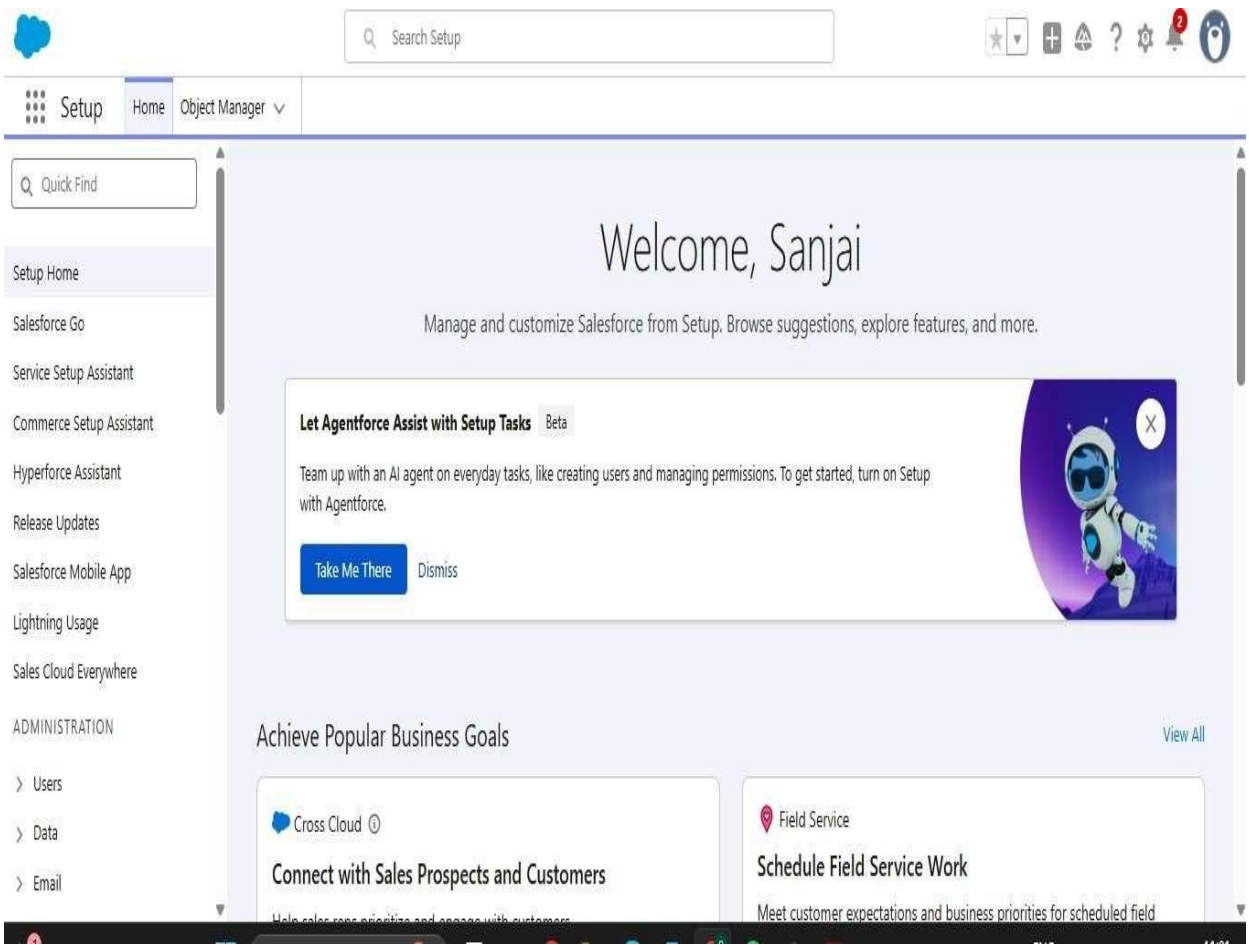
Salesforceisyourcustomersuccessplatform,designedtohelpyousell,service,market, analyze, and connect with your customers. Salesforcehaseverythingyouneedtorunyourbusinessfromanywhere.Usingstandard products and features, you can manage relationships with prospects and customers, collaborateandengagewithemployeesandpartners,andstoreyourdatasecurelyinthe cloud.

Activity1:
CreatingaSalesforceDeveloperOrg:

ADeveloperorghasallthefeaturesandlicensesyouneedtogetstartedwithSalesforce.

1. SearchDeveloper.salesforce.com/signup
2. EnterthefollowingdetailslikeFirstname,lastname,Email,Role,Company, Country/Region, Postal code, and Username must be unique.
3. Clicksignmeup,afterafewminyouwillreserveamailsalesforceorgandbyusing the verify account link you can create your new password.
4. Clicksave.
5. Searchlogin.salesforce.com

ByusingusernameandpasswordyoucanintotheSalesforceorg.Thehomepagewill appear below

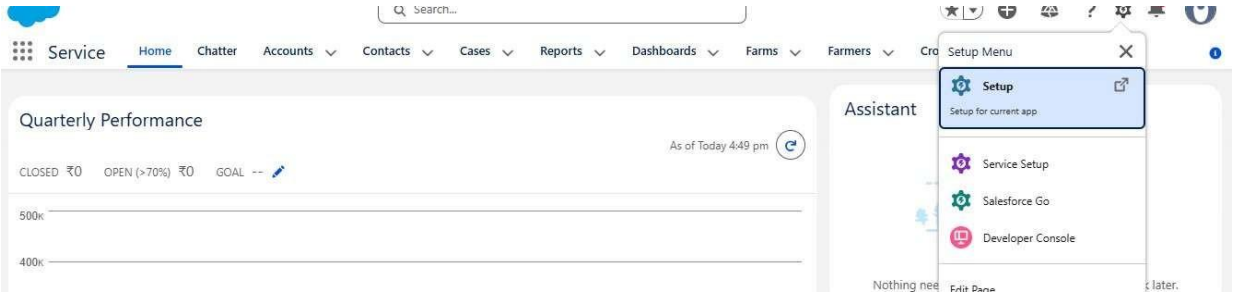


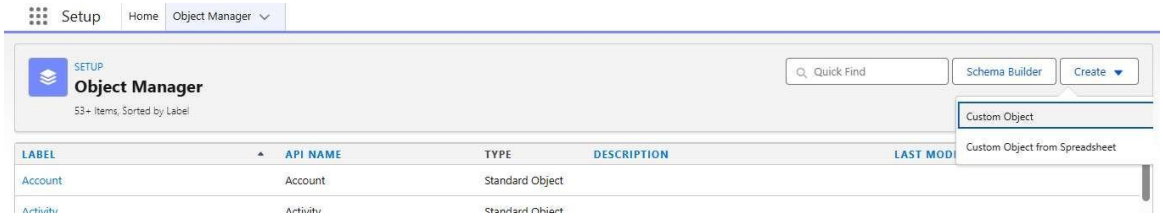
Milestone2-Object:

Whatisan object?
Salesforceobjectsaredatabasetablesthatpermityoutostoredatathatisspecifictoan organization. It consists of fields (columns) and records (rows).

InThisApplicationWeUse4CustomObjects:
1.Farm
2.Crop
3.Farmer
4.Buyer

Activity1:
CreateacustomobjectforFarm:
Tocreateacustomobject,followthesesteps:
1. Fromsetupclickonobjectmanager.
2. Clickcreate,selectcustomobject.





3. Fillinthelabelas"Farm".
4. Fillintheplurallabelas"Farms".
5. Recordname:FarmName"
6. Selectthedatatypeas"Text".

7. IntheOptionalFeaturessection,selectAllowReportsandTrackFieldHistory.
8. IntheDeploymentStatussection,ensureDeployedisselected.
9. IntheSearchStatussection,selectAllowSearch.
10. IntheObjectCreationOptionssection,selectAddNotesandAttachmentsrelatedlist to default page layout.

11. Leaveeverythingelseasis,andclickSave.

Activity2:

CreationofCropobject:
1. ClickonthegeariconandthenselectSetup.
2. Clickontheobjectmanagertabjustbesidethehometab.
3. Aftertheabovesteps,havealookontheextremerightyouwillfindaCreate Dropdown click on that and select Custom Object.
a. OntheCustomObjectDefinitionpage,createtheobjectasfollows:
b. Label:Crop
c. PluralLabel:Crops
d. RecordName:CropName

e. Selectthedatatypeas"Text".
f. CheckktheAllowReportscheckboxg.CheckktheAllowSearchcheckbox
h.IntheObjectCreationOptionssection,selectAddNotesandAttachmentsrelatedlist to default page layout i. Click Save.

Activity3:

CreationofFarmerobject:
1. ClickonthegeariconandthenselectSetup.
2. Clickontheobjectmanagertabjustbesidethehometab.
3. Aftertheabovesteps,havealookontheextremerightyouwillfindaCreate Dropdown click on that and select Custom Object.
a. OntheCustomObjectDefinitionpage,createtheobjectasfollows:
b. Label:Farmer

c. PluralLabel:Farmers
d. RecordName:FarmerName e.Selectthedatatypeas"Text".
f.Checkthe AllowReportscheckbox.g.Checkthe AllowSearchcheckbox.
h.IntheObjectCreationOptionssection,selectAddNotesandAttachmentsrelatedlist to default page layout. i. Click Save.

Activity4:

CreationofBuyerobject:
1. Clickonthegeariconandthenselect Setup.
2. Clickontheobjectmanagertabjustbesidethehometab.
3. Aftertheabovesteps,havealookontheextremerightyouwillfindaCreate Dropdown click on that and select Custom Object.
a. OntheCustomObjectDefinitionpage,createtheobjectasfollows:
b. Label:Buyer
c. PluralLabel:Buyers
d. RecordName:BuyerName
e. Selectthedatatypeas"Text".
f. Checkthe AllowReportscheckbox.g.Checkthe AllowSearchcheckbox.
h. IntheObjectCreationOptionssection,selectAddNotesandAttachmentsrelatedlist to default page layout.
i. ClickSave.

Milestone3-
Tab: Tab

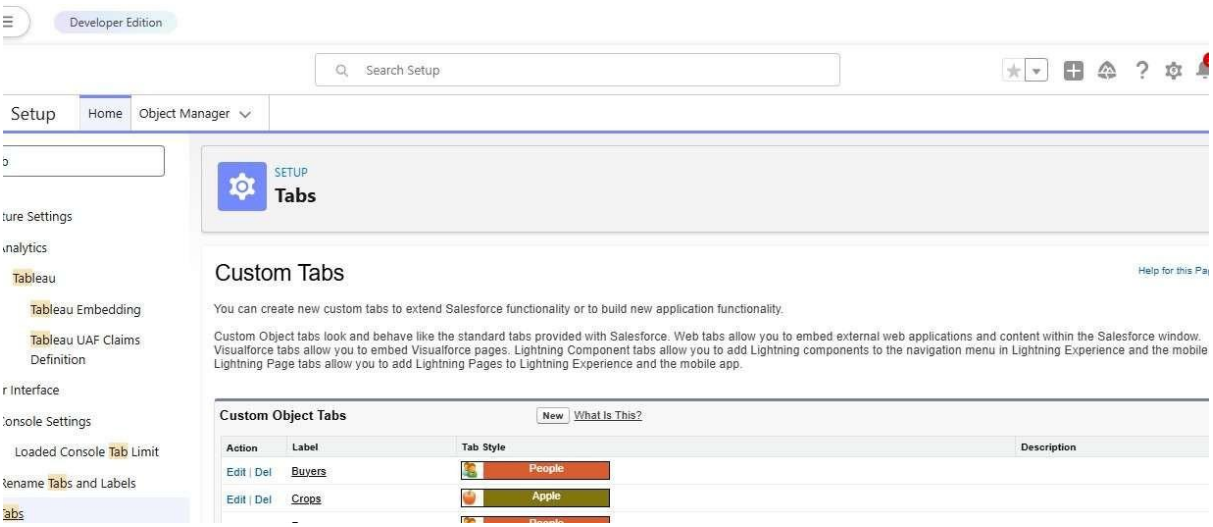
TabsinSalesforcehelpusersviewtheinformationataglance.Itdisplaysthedataof objects and other web content in the application.

Activity1:

CreationofFarmtab:

Nowcreateacustomtab.

1. ClickontheHometab,enterTabsinQuickFindandselectTabs.



2. Undercustomobjecttabs,clickNew.

3. ForObject,selectFarm.

ForTabStyle,selectanyicon.

4. Leavealldefaultsasis.ClickNext,Next,andSave.

Activity2:

CreationofCroptab:

Nowcreateacustomtab.

1. ClickonHometab,enterTabsinQuickFindandselectTabs.
2. Undercustomobjecttabs,clickNew.
3. ForObject,selectCrop.

4. ForTabStyle,selectanyicon.
5. Leavealldefaultsasis.ClickNext,Next,andSave.

Activity3:

CreationofFarmertab:

Nowcreateacustomtab.

1. ClickontheHometab,enterTabsinQuickFindandselectTabs.
2. Undercustomobjecttabs,clickNew.
3. ForObject,selectFarmer.
4. ForTabStyle,selectanyicon.
5. Leavealldefaultsasis.ClickNext,Next,andSave.

Activity4:

CreationofBuyertab:

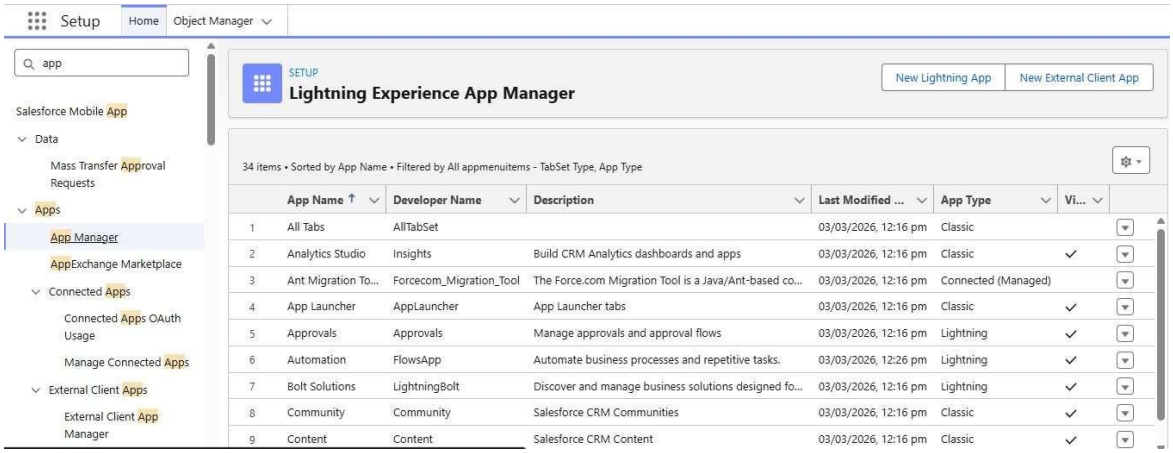
Nowcreateacustomtab.

1. ClickonHometab,enterTabsinQuickFindandselectTabs.
2. Undercustomobjecttabs,clickNew.
3. ForObject,selectBuyer
4. ForTabStyle,selectanyicon.
5. Leavealldefaultsasis.ClickNext,Next,andSave.

Milestone4-LightningApp:

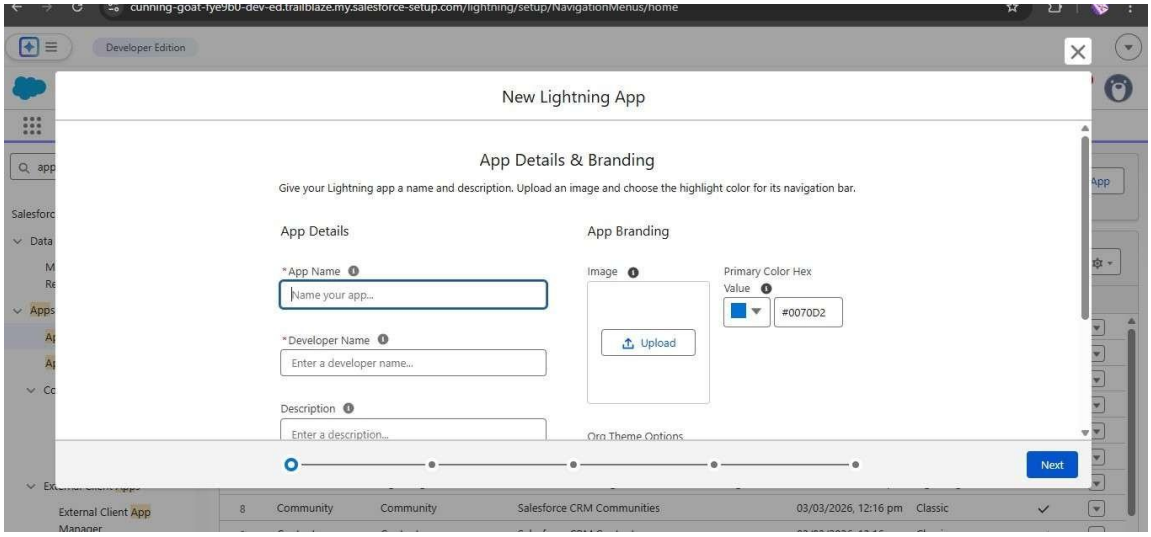
WhatisanApp?
Apps in Salesforce area group of tabs that help the application function by working togetherasaunit.Ithasaname,alogo,andaparticularsetoftabs.Thesimplestapp usually has justtwo tabs.

Activity1:
CreatetheFarmManagementSystemapp

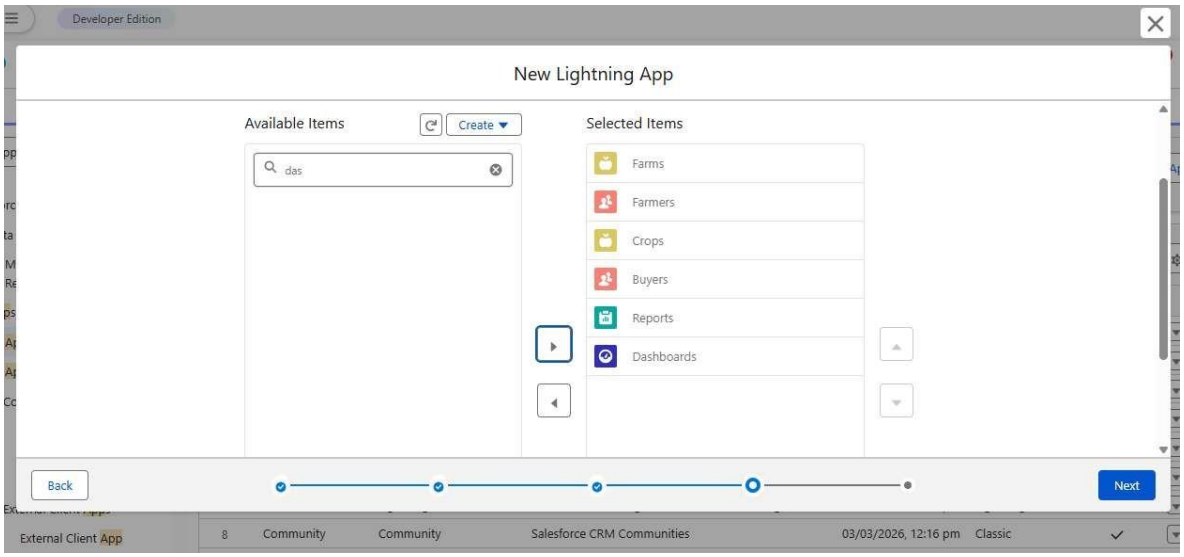


● FromSetup,enterAppManagerintheQuickFindandselectAppManager.

● ClickNewLightningApp.FarmManagementSystemastheAppName,thenclick Next
● UnderAppOptions,leavethedefaultselectionsandclickNext.
● UnderUtilityItems,leaveasisandclickNext.
● FromAvailableItems,selectFarm,Crop,Farmer,Buyer Reports,andDashboardsandmovethemtoSelectedItems.ClickNext.●FromAvailable Profiles, select System Administrator and move it toSelected Profiles.Click Save &Finish.
● Toverifyyourchanges,clicktheAppLauncher,typeFarmManagementSystemand select the Job Application app.
Note:
1. AppLauncher-Displaysavailableapps.
2. AppName-Displaysthecurrentselectedapp.



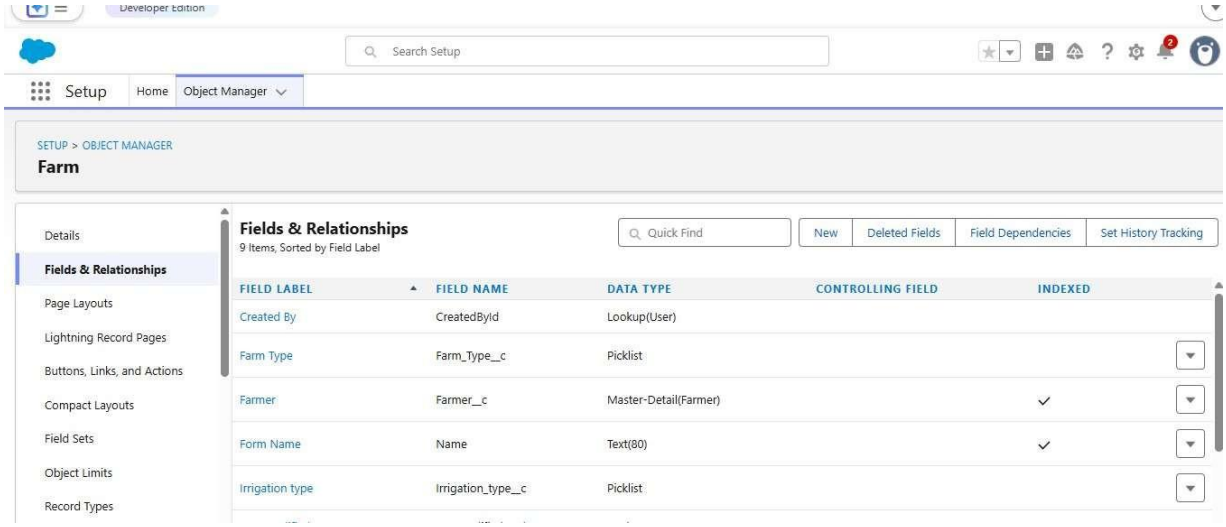
3. Navigationmenu-Displaysthetabsavailableinsidetheapp.



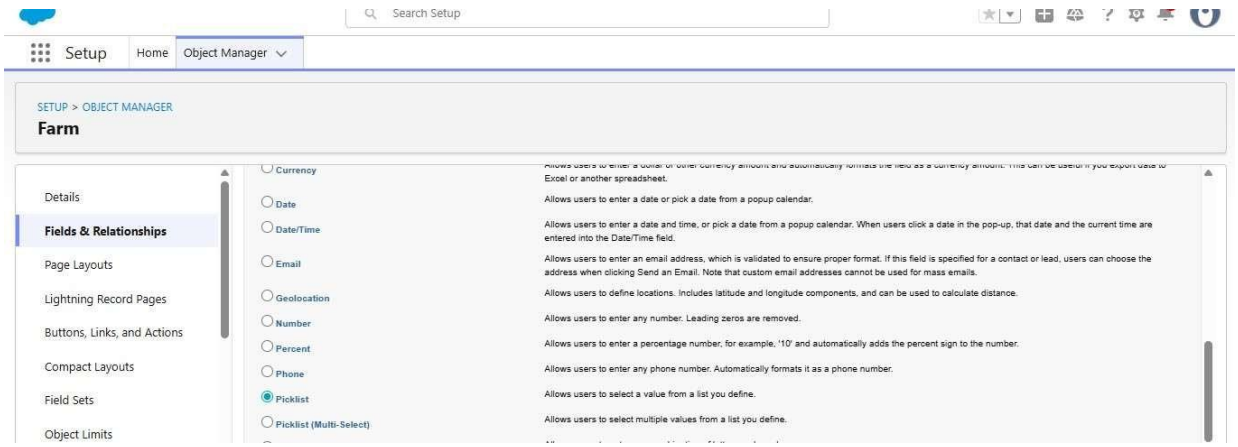
Milestone5-
FieldsandRelationship:Whatarefields?

FieldsinSalesforcerepresentwhatthecolumnsrepresentinrelationaldatabases.Itcan store data values which are required for a particular object in a record.

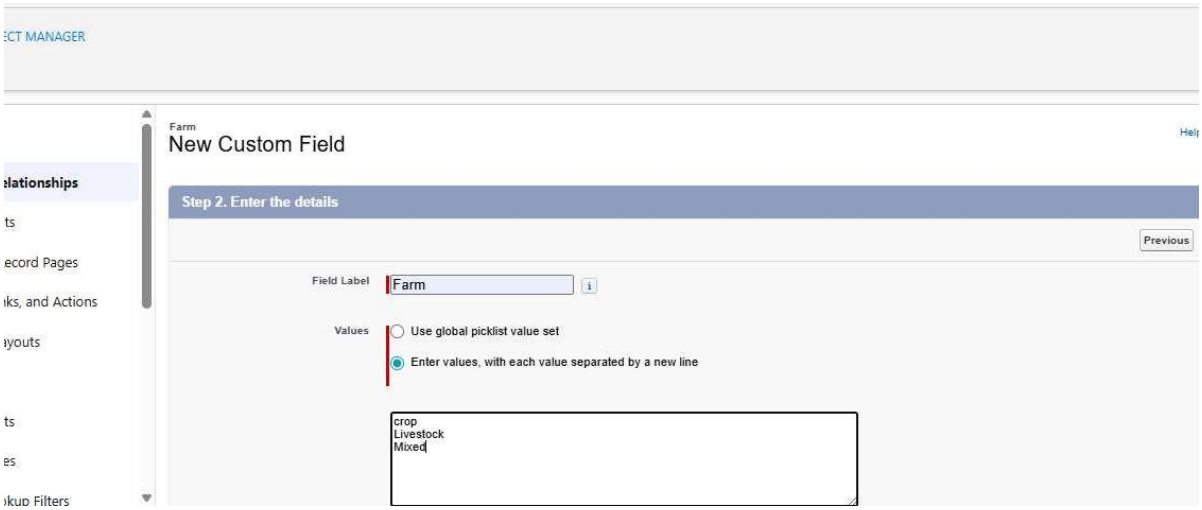
Activity1:
CreationoffieldsfortheFarmobjects:
1. clickthegeariconandselectSetup.ThislaunchesSetupinanew tab.
2. ClicktheObjectManagertabnexttoHome.
3. SelectFarm.
4. SelectFields&Relationshipsfromtheleftnavigation,andclickNew



5. ChooseDatatypeasapicklist,Clicknext



6. EnterFieldLabel(FarmType),andEnterValuesas(Crop,Livestock,Mixed
) 7.Next,Next Click Save.



Nowlet'screatetheotherfieldsfollowingActivity1steps1to4andwemustchoosethe data types of the fields carefully.

1. NextweCreateanotherPicklistField,Enterfieldlabel(SoilQuality)entervalues (Organic,Cover Crops,Composting)
2. AgainweCreatePicklistfield,FieldLabel(Irrigationtype),EnterValues(Sprinkler irrigation,Drip irrigation,localized irrigation)
3. Choosedatatypeasatext,enterfieldlabel(Location)4.Choosedatatypeastext,enter field label (size)
5. CreateaMasterDetailrelationShipwithFarmerObject.

Activity2:
CreationoffieldsfortheCropobjects:

1.ChooseDatatypeAsatext,EnterFieldLabelname(CropName),Nextsave.2.Select Data type as date ,Enter Field label (Planting Date),then save .
3.Againweselectdatatypeasadate,Enterfieldlabe(Harvestdate),save
4.NextWeCreateLookuprelationshiponcropobjectrealtedtoFarmObject.
5.Weselectadatatypeascurrency,EnterfieldLabel(ActualYield),NextSaveit.
6. SelectDatatypeasaNumber,enterFieldLabel(TotalAcresHarvested),Enter Maximum Length(20) and save.
7.NowWecreateFormulaFieldForThatSelectdataTypeAsaFormula,ClickNext.

Setup | Home | Object Manager

SETUP > OBJECT MANAGER

Crop

- Details
- **Fields & Relationships**
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types

Data Type

None Selected	Select one of the data types below.
Auto Number	A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.
Formula	A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.
Roll-Up Summary	A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.
Lookup Relationship	Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.
Master-Detail Relationship	Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where: • The relationship field is required on all detail records. • The ownership and sharing of a detail record are determined by the master record. • When a user deletes the master record, all detail records are deleted. • You can create rollup summary fields on the master record to summarize the detail records. The relationship field allows users to click on a lookup icon to select a value from a popup list. The master object is the source of the values in the list.
External Lookup Relationship	Creates a relationship that links this object to an external object whose data is stored outside the Salesforce org.

8. EnterFieldLabel(ExpectedYieldperAcre),Selectformulareturntypeas Currency,click Next.

9. EnteraFormulaActual,thenclicknextand save.

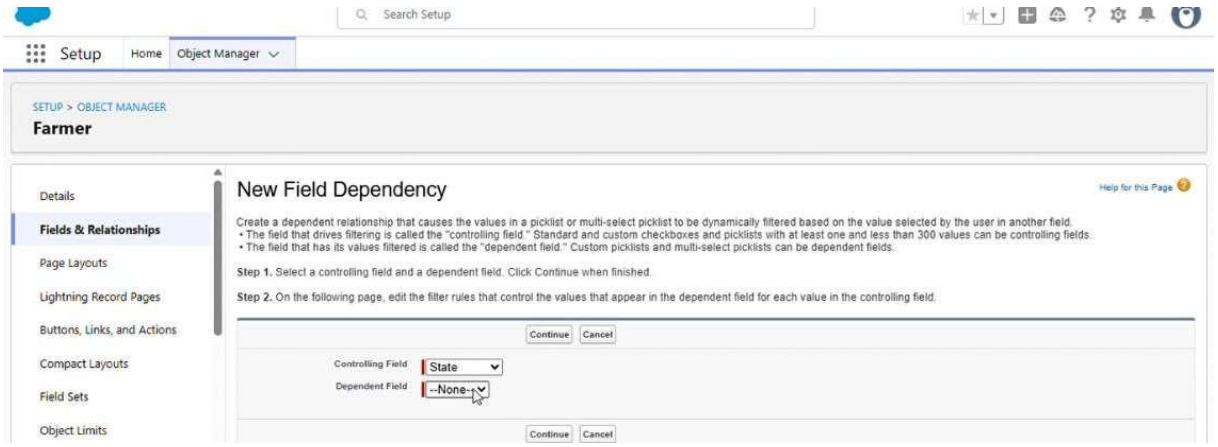
Activity3:
CreationoffieldsfortheFarmerobjects:

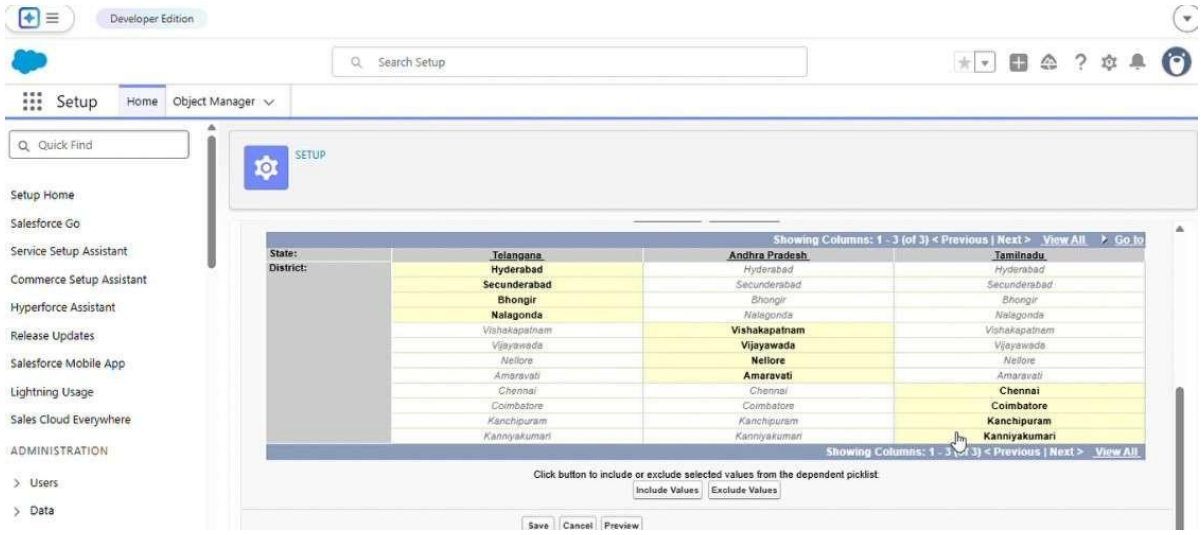
1. SelectDatatypePhone,Enterfieldlabel(ContactNumber),nextSave.
2. SelectDatatypePicklist,EnterFieldLabel(State),Entervalues(Telangana,Andhra Pradesh,Tamilnadu),save.
3. ChooseDatatypePicklist,EnterFieldLabel(District),EnterValuesofdistrict,nextsave.
4. AgainselectDatatypeasPicklist,EnterFieldLabel(Village),EnterValuesof villages,next save.

Creationoffielddependency:

1. NowcreateFieldDependencyForDistrictfield,OpenObjectManager,ChooseFarmer object ,select Fields&Relationships,On Right side We have Field Dependency option,select that option.
2. Clicknew,SelectControllingFieldasState,DependentFieldAsDistrict,Clickon Continue.
3. SelecttheappropriateValueineachcolumnbydouble-clickingthem.
4. ForEx.Telangana-YadadriBhongir.
5. ClickIncludeValues.AnditisalsothesameforAndhraPradesh,Tamilnaduwithits city.

6. ClickPreview,thentestthedependencybyselectingadifferentStateandviewingthe associate Values available for Particular state.


Search Setup
Setup
Home
Object Manager
SETUP > OBJECT MANAGER
Farmer
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
New Field Dependency
Help for this Page
Create a dependent relationship that causes the values in a picklist or multi-select picklist to be dynamically filtered based on the value selected by the user in another field.
• The field that drives filtering is called the "controlling field." Standard and custom checkboxes and picklists with at least one and less than 300 values can be controlling fields.
• The field that has its values filtered is called the "dependent field." Custom picklists and multi-select picklists can be dependent fields.
Step 1. Select a controlling field and a dependent field. Click Continue when finished.
Step 2. On the following page, edit the filter rules that control the values that appear in the dependent field for each value in the controlling field.
Continue
Cancel
Controlling Field
State
Dependent Field
--None--
Continue
Cancel


Developer Edition
Search Setup
Setup
Home
Object Manager
Quick Find
Setup Home
Salesforce Go
Service Setup Assistant
Commerce Setup Assistant
Hyperforce Assistant
Release Updates
Salesforce Mobile App
Lightning Usage
Sales Cloud Everywhere
ADMINISTRATION
Users
Data
SETUP
Showing Columns: 1 - 3 (of 3) < Previous | Next > View All > Go to
State:
District:
Telangana
Andhra Pradesh
Tamilnadu
Hyderabad
Secunderabad
Bhongir
Nalagonda
Vishakapatnam
Vijayawada
Nellore
Amaravati
Chennai
Coimbatore
Kanchipuram
Kanniyakumari
Showing Columns: 1 - 3 (of 3) < Previous | Next > View All
Click button to include or exclude selected values from the dependent picklist.
Include Values
Exclude Values
Save
Cancel
Preview

7. ClickClosetoclosethepreviewwindow,ClickSave.

Activity4:

CreationoffieldsfortheBuyerobjects:

1.SelectDatatypeasaPhone,EnteraFieldLabel(Contact),Save.
2.SelectDatatypeasanEmail,EnterFieldLabel(Email),Nextsave.
3.SelectDatatypeTextarea,EnterFieldLabel(Address),Saveit.
4. SelectDatatypePicklist,FieldLabel(BuyerType),Entervalu
es (Wholesale,Retail),Click Save.

Milestone6-Profile:

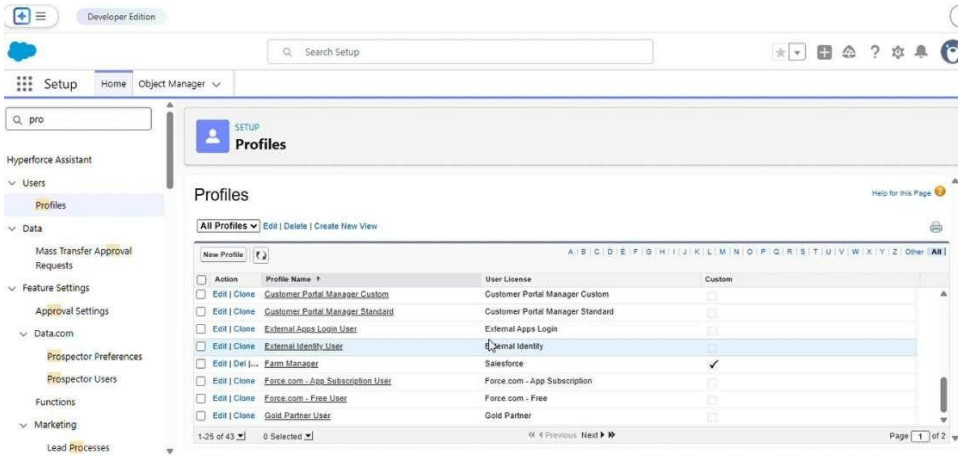
Whatisa profile?
Aprofileisagroup/collectionofsettingsandpermissionsthatdefinewhatausercando in

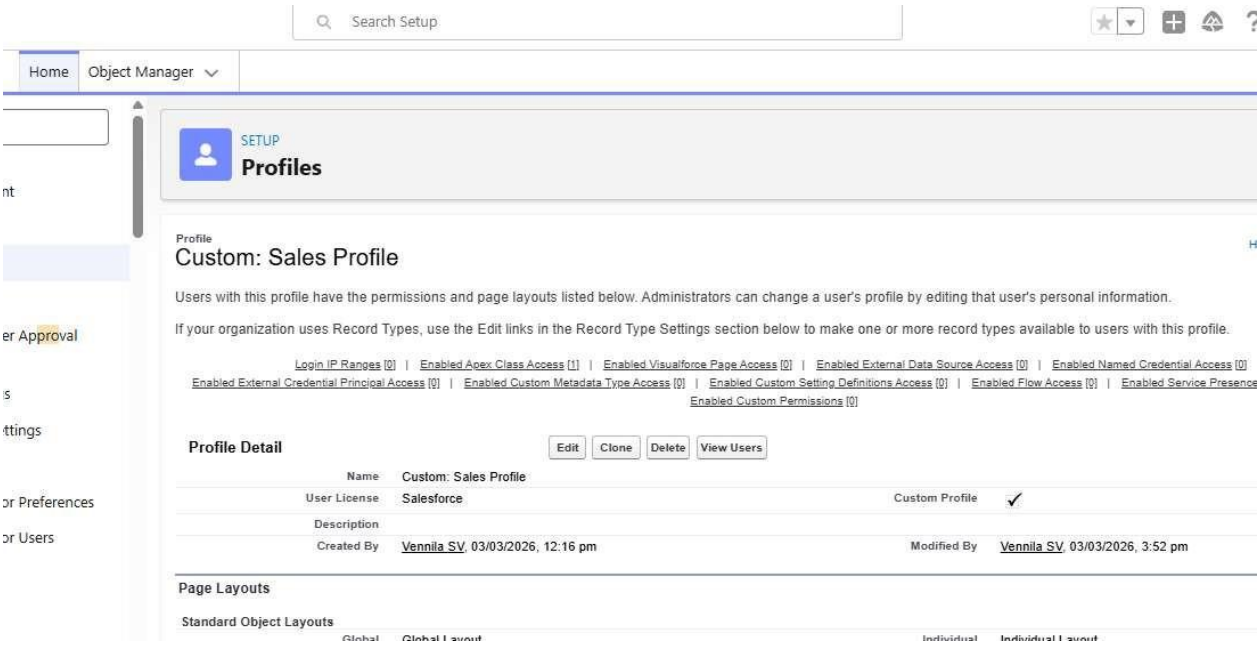
salesforce.Aprofilecontrols“Objectpermissions,Fieldpermissions,Userpermissions,
Tab
settings,Appsettings,Apexclassaccess,Visualforcepageaccess,Pagelayouts,Record Types,
Login hours & Login IP ranges.
Aprofilecanbeassignedtomanyusers,butusercanbeassignedsingleprofileatatime.

Activity1:

Createacustomprofile:

1. Fromsetup,enterprofilesinQuickFind box
2. Selectprofiles(Standarduser).
3. Clickclone.
4. ForProfile,enterSalesDepartment.


Developer Edition
Search Setup
Setup
Home
Object Manager
pro
Hyperforce Assistant
Users
Profiles
Data
Mass Transfer Approval Requests
Feature Settings
Approval Settings
Data.com
Prospector Preferences
Prospector Users
Functions
Marketing
Lead Processes
SETUP
Profiles
Profiles
Help for this Page
All Profiles
Edit | Delete | Create New View
New Profile
A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other | All
Action
Profile Name
User License
Custom
Edit | Clone
Customer Portal Manager Custom
Customer Portal Manager Custom
Edit | Clone
Customer Portal Manager Standard
Customer Portal Manager Standard
Edit | Clone
External Apps Login User
External Apps Login
Edit | Clone
External Identity User
External Identity
Edit | Del |...
Farm Manager
Salesforce
Edit | Clone
Force.com - App Subscription User
Force.com - App Subscription
Edit | Clone
Force.com - Free User
Force.com - Free
Edit | Clone
Gold Partner User
Gold Partner
1-25 of 43
0 Selected
Previous
Next
Page 1 of 2


Search Setup
Home
Object Manager
SETUP
Profiles
Profile
Custom: Sales Profile
Users with this profile have the permissions and page layouts listed below. Administrators can change a user's profile by editing that user's personal information.
If your organization uses Record Types, use the Edit links in the Record Type Settings section below to make one or more record types available to users with this profile.
Login IP Ranges [0] | Enabled Apex Class Access [1] | Enabled Visualforce Page Access [0] | Enabled External Data Source Access [0] | Enabled Named Credential Access [0]
Enabled External Credential Principal Access [0] | Enabled Custom Metadata Type Access [0] | Enabled Custom Setting Definitions Access [0] | Enabled Flow Access [0] | Enabled Service Presence
Enabled Custom Permissions [0]
Profile Detail
Edit
Clone
Delete
View Users
Name
Custom: Sales Profile
User License
Salesforce
Custom Profile
Description
Created By
Vennila SV, 03/03/2026, 12:16 pm
Modified By
Vennila SV, 03/03/2026, 3:52 pm
Page Layouts
Standard Object Layouts

5. Clicksave.

Activity2:

Createacustomprofile-2:
Createaprofilewiththeprofilenameas“FarmManager”.

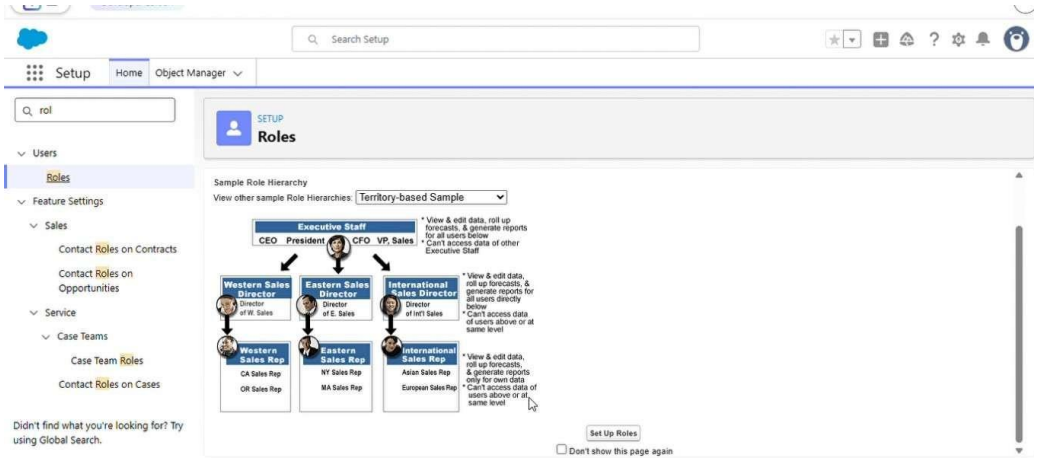
6. Fromsetup,enterprofilesinQuickFind box
7. Selectprofiles(Standarduser).
8. Clickclone.

Milestone7-Role

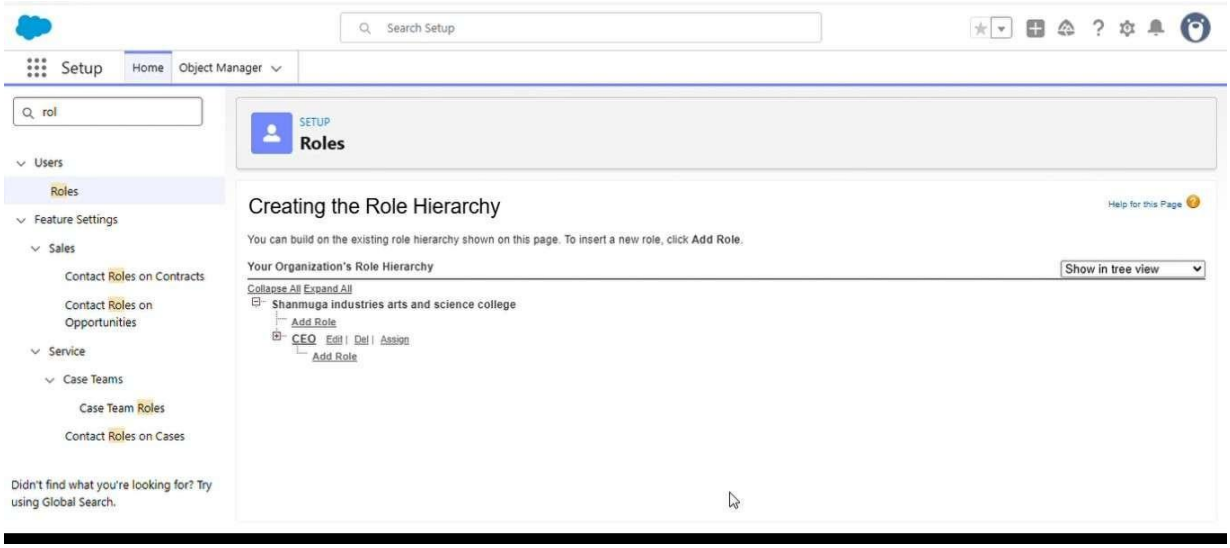
In Salesforce, roles are used to determine which users have access to certain data and functionswithinthesystem.Theyarealsousedtodefinethereportinghierarchywithin an organization.Users with higher roles have greater access to data and more control over the system.

Activity-1
Creationofrole

1. FromtheQuickfindboxsearchfortheroleandclickontherolesoption
2. selectthesetuprolesoption



3. BelowtheCEOclickonaddroleandenterthelabelnameasa”FarmManager” and role name will be Automatically populated and click on save.



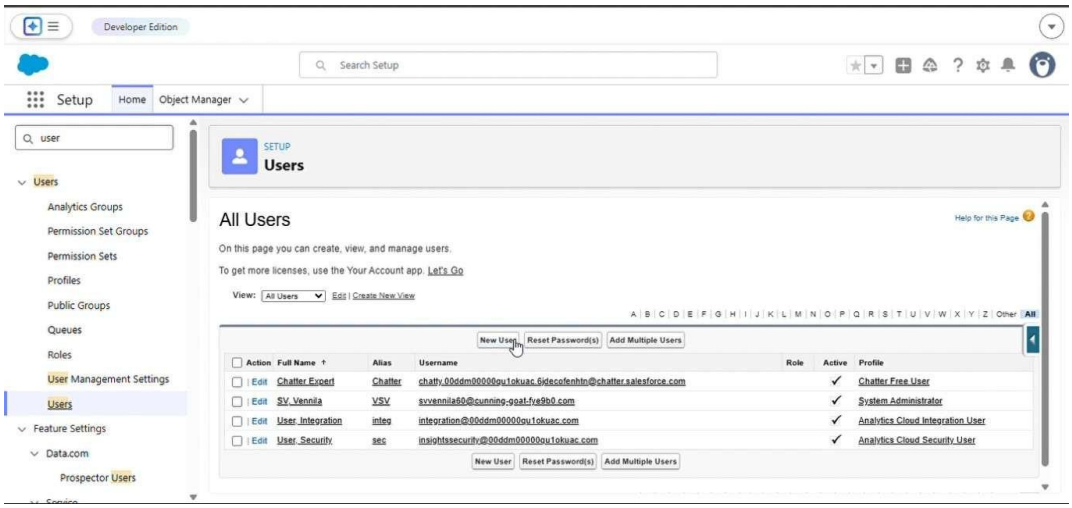
Milestone8-User:

Whatisa user?
Auserisanyonewhologsintoalesforce.Usersareemployeesatyourcompany,suchas salesreps,managers,andITspecialists,whoneedaccesstothecompany'srecords.Every user in Salesforcehasauseraccount.Theuseraccountidentifiestheuser,andtheuseraccount settings determine what features and records the user can access.

NOTE-AsSalesforcelicensecanonlybeusedby2UsersatatimeinDevOrg,soIfyou don't find salesforcelicensethendeactivateauserwhohassalesforcelicenseOrchangethelicense type from Salesforce to any other.

Activity1:

ToCreateauser:
1. FromSetup,enterUsersintheQuickFindbox,thenselectUsers.
2. ClickNewUser.



3. EnterFirstnameasFarmandlastnameasManager.
4. Entertheuser'snameandemailaddressandauniqueusernameintheformofanemail address. By default, the username is the same as the email address.
5. SelecttheroleFarmManager.
6. SelectuserLicenseasStandardPlatformUser.
7. Selectprofile(FarmManager).
8. Clicksave.

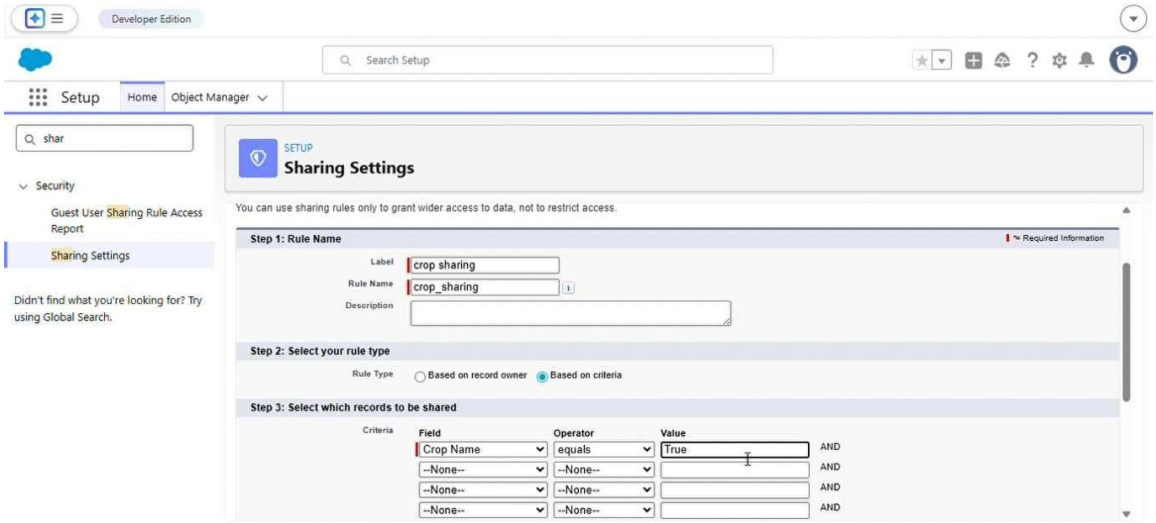
Activity2:
Createauserwithausernameas"SalesDepartment",andassignhimthesalesManager profile.
Follow the steps from above Activity

Milestone9-SharingRules:

WhatareSharingRules?
Sharing rules help users to share records based on conditions. It is basically created for objectswhoseorganization-widedefaults(OWD)aresettopublicread-onlyorprivate because sharing rules can only extend the access and not restrict it.
Typesofsharingrules,
• Owner-basedSharingRules•Criteria-basedSharingRules

Activity1:

Createasharingrule
1) GotoSharingSettings,whichcanbefoundundertheQuickFindsection.
2) Scrolldownandfindthecandidateobjectwhereasharingruleneedstobeadded,and then click on New to create a new sharing rule.
3) Addthelabelofthesharingruleyouwanttomake.
4) Selectyourruletypebasedonthecriteria.
5) Selectthefieldcanjoinimmediatelycheckfieldfromthe Cropobject.
6) SelecttheCropasequalandvalueisTrue.
7) Andinselectingtheuserstosharewiththesectionselectrolesandinthatselect Farm Manager.
8) Andinthesectionofselectthelevelofaccessfortheusersgivetheaccess Read/Write.
9) Andsavetherule.

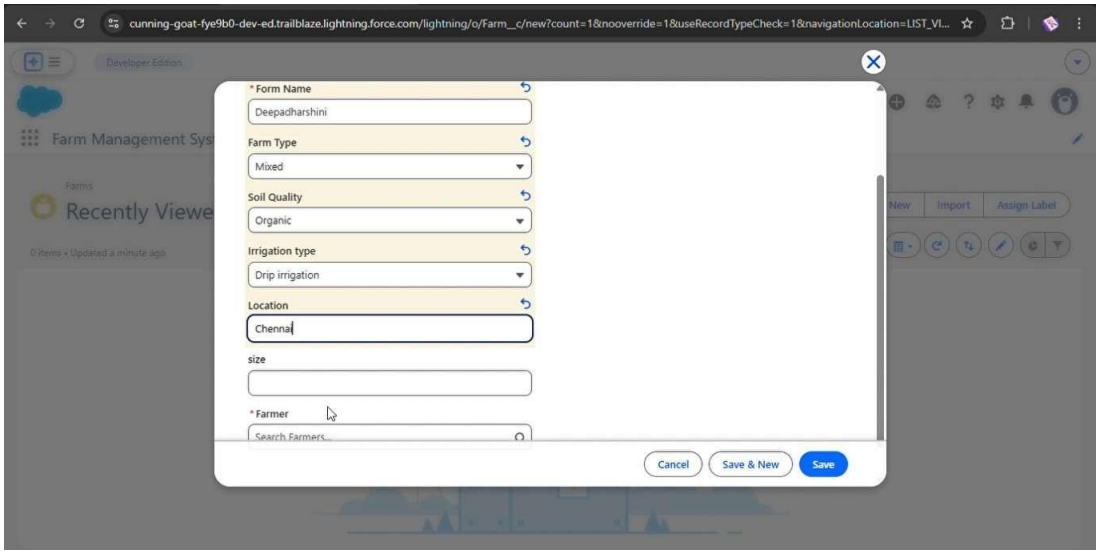

Developer Edition
Search Setup
Setup
Home
Object Manager
shar
Security
Guest User Sharing Rule Access Report
Sharing Settings
Didn't find what you're looking for? Try using Global Search.
SETUP
Sharing Settings
You can use sharing rules only to grant wider access to data, not to restrict access.
Step 1: Rule Name
Required Information
Label
crop sharing
Rule Name
crop_sharing
Description
Step 2: Select your rule type
Rule Type
Based on record owner
Based on criteria
Step 3: Select which records to be shared
Criteria
Field
Operator
Value
Crop Name
equals
True
AND
--None--
--None--
AND
--None--
--None--
AND
--None--
--None--
AND

Activity2:

CreateaSharingRuletoSharetherecordsofBuyertoSalesdepartmentwiththeAccess of Read/Write.
FollowthestepsfromaboveActivity.

Milestone-10: User Adoption
Activity1:CreateaRecord(Farmer)

- ClickonAppLauncheronleftsideofscreen.
- SearchFarmManagement&clickon it.
- ClickonFarmer Tab.
- Clicknewandfilldetails&Save.



Milestone 11-Reports:

WhatareReports?

A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is storedinafolder.Folderscanbepublic,hidden,orshared,andcanbesettoread-onlyor read/write.

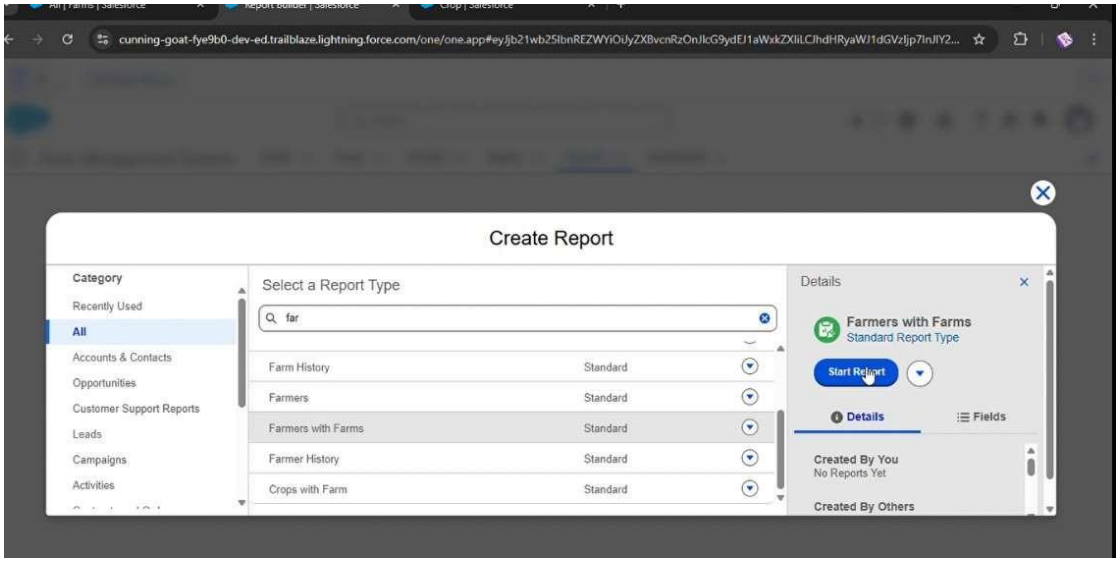
Activity1:

Createareport:

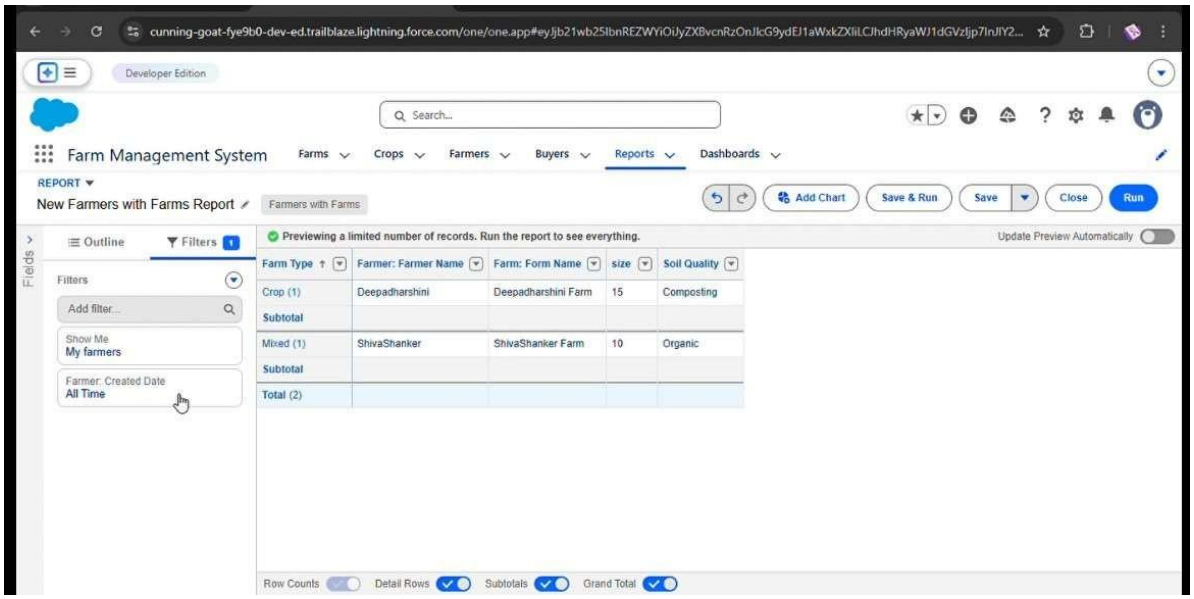
Createareportthatdisplaysratingoftheaccountandwhichhastypeandaccountname.

1) Clickonapplauncher searchforreports.

2) Clickonthenewreportandselectthecategoryhas FarmerwithFarms.



3) Inthedetailssectionselecttheoptionstartreport.
4) InOutlineClickonColumnSelectWhichFieldyouwanttodisplayintheReport.
5) ClickGroup,SelectWhichFieldsYouWanttoGroup.
6) Savethereportbygivinglabelname(FarmerwithFarms)andsavethefolderasa public folder and save the report.

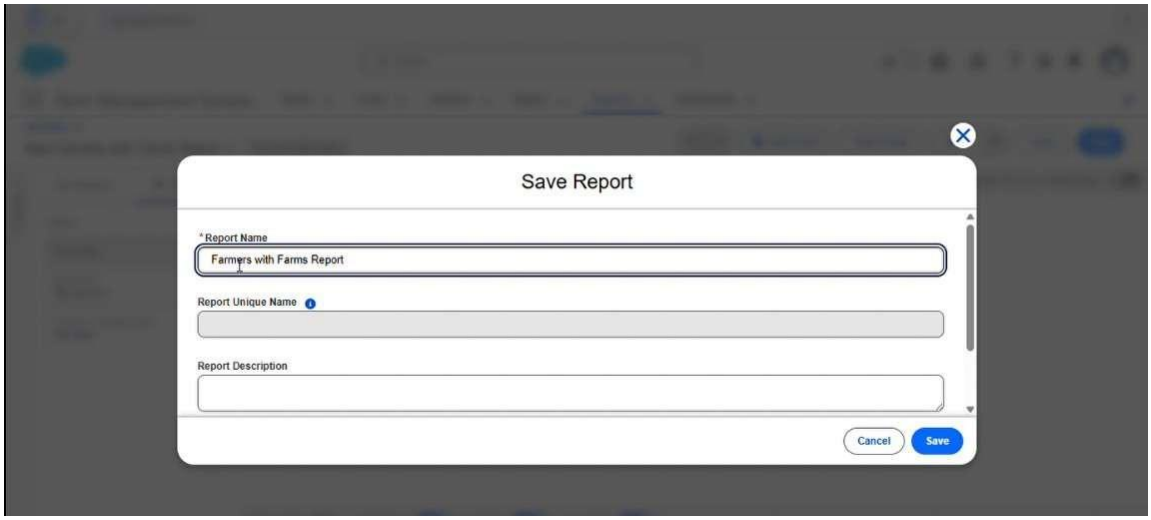


Activity2:
CreateaReportusingtheObjectsCrops,Buyers.FollowthestepsfromaboveActivity.

Milestone12-Dashboard:

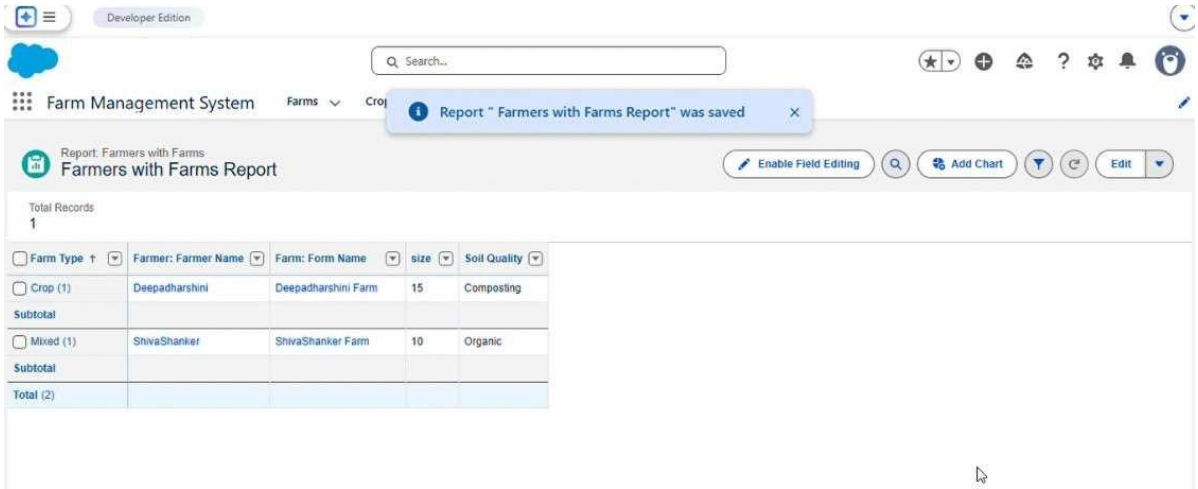
Dashboards provide more insights than reports as they combine the data from many reports andshow a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. Thedashboardscancontaincharts,graphsand Tabular data.

Activity1:
CreateaDashboard:
1. ClicktheDashboardtab.
2. ClickNewDashboard.
3. NamethedashboardFarmerswithfarmsandclickCreate.
4. Click+Component.
5. SelecttheFarmerswithFarmsandclickSelect.
6. SelecttheGaugechartandclickAdd.
7. ClickSaveandthen Done.



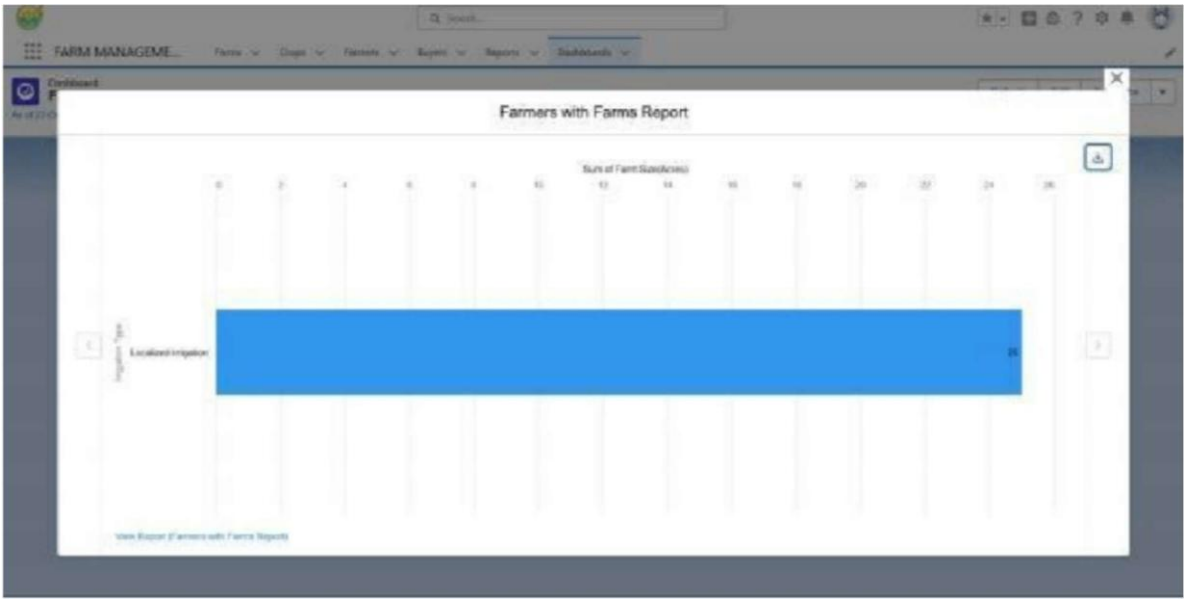
Milestone-13:ViewReportandDashboard:
Activity 1: Report

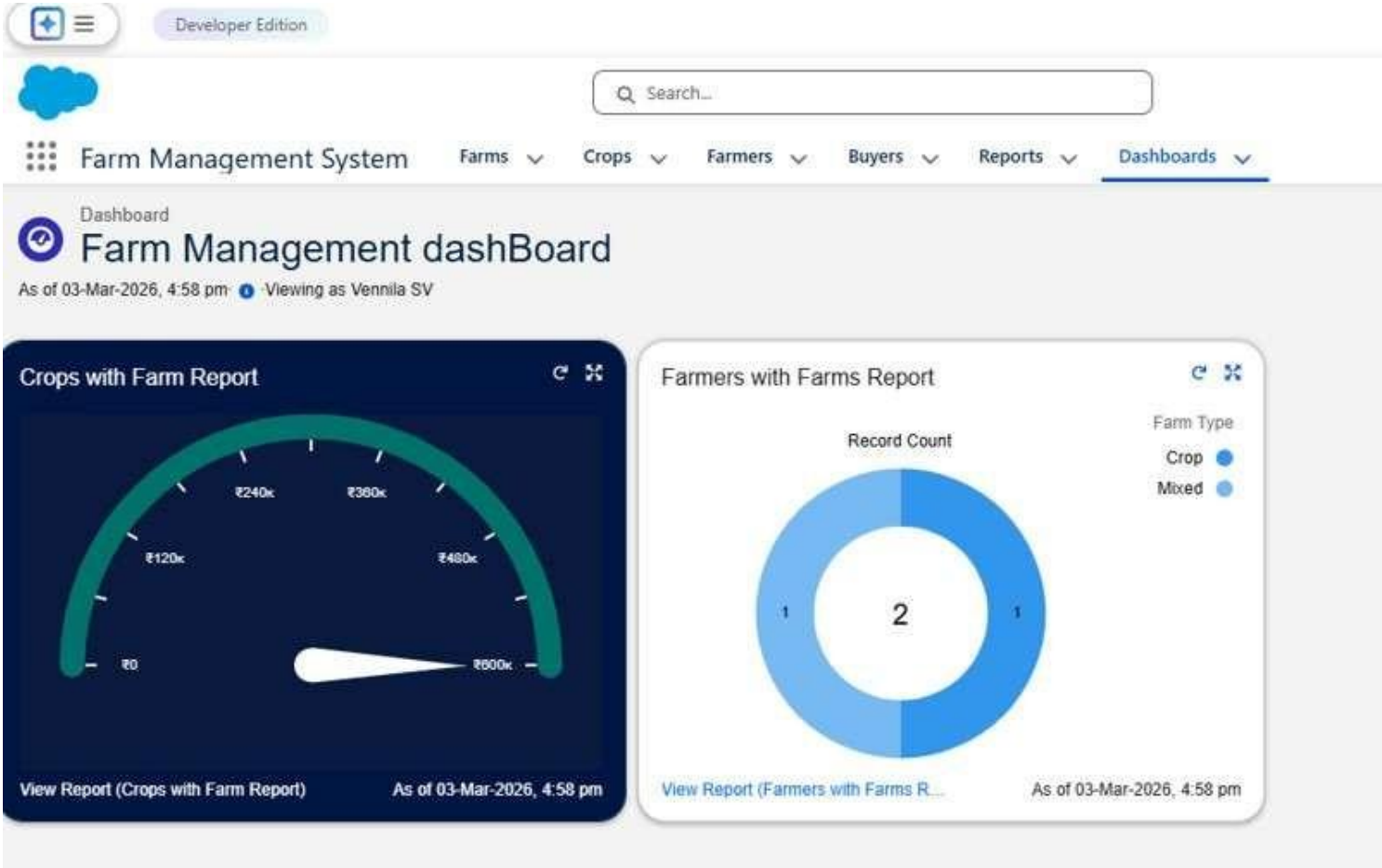
- ClickonAppLauncheronleftsideofscreen.
- SearchFarmManagementSystem&clickon it.
- ClickonReportsTab.
- ClickonFarmerswithFarms&seerecords



Activity2:Dashboard

- ClickonAppLauncheronleftsideofscreen.
- SearchFarmManagementsystem&clickon it.
- ClickonDashboardTab.
- ClickFarmersWithfarms&seerecords.




Developer Edition
Search...
Farm Management System
Farms
Crops
Farmers
Buyers
Reports
Dashboards
Dashboard
Farm Management dashBoard
As of 03-Mar-2026, 4:58 pm · Viewing as Vennila SV
Crops with Farm Report
₹0
₹120k
₹240k
₹360k
₹480k
₹600k
View Report (Crops with Farm Report)
As of 03-Mar-2026, 4:58 pm
Farmers with Farms Report
Record Count
Farm Type
Crop
Mixed
1
2
1
View Report (Farmers with Farms R...
As of 03-Mar-2026, 4:58 pm

7. FUNCTIONAL&PERFORMANCETESTING

Duringthetestingphase,screenshotswerecapturedfromtheSalesforceinterfacetoverifysystem functionality. The following were tested:

- CreationofFarm,Crop,Farmer,andBuyerrecords
- Validationofinputfields
- Relationshipanddependencychecks
- Profileandroleaccesstesting
- Sharingruleverification
- Reportanddashboardgeneration

Performancetestingensuredsmoothsystemresponseandstabilityduringmultipleuser operations.

8. ADVANTAGES&DISADVANTAGES

Advantages

- Centralizeddatasystem
- Reducedpaperwork
- Improvedaccuracy
- Bettermonitoring
- Real-timeanalytics

Disadvantages

- Requiresinternetconnection
- Salesforcelicensedependency
- Limitedofflineaccess

9. Conclusion:

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

10. FUTURESCOPE

- Mobileapplicationintegration
- AI-basedcropprediction
- Weatherdataintegration
- IoTsensorconnectivity
- Governmentschemeintegration
- Marketpriceforecasting

11. APPENDIX

- SourceCode:SalesforceConfigurationandMetadata
- Dataset:Farm,Crop,Farmer,andBuyerRecords
- ToolsUsed:SalesforceDeveloperOrg

ThankYou